



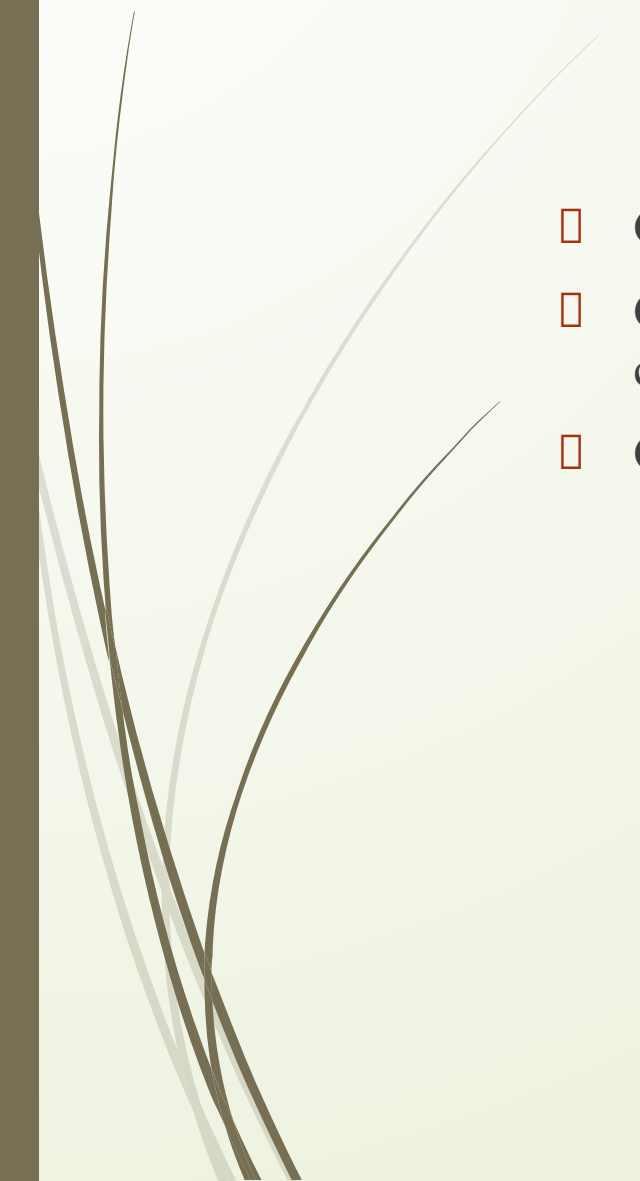
Business Process Engineering

Sobia.Iftikhar@nu.edu.pk

Week 05



Control Flow Patterns

- ❑ Control flow patterns provide a bench mark/scale for expressing process orchestrations.
 - ❑ Control flow patterns are independent of concrete process languages, so that each pattern can be expressed in different process languages.
 - ❑ Control flow patterns can also be used to compare the expressiveness of process languages.
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control flow patterns

- Basic control flow patterns include
 - sequence,
 - and split,
 - and join,
 - exclusive or split
 - exclusive or join

Sequence Control Flow Pattern

- A task in a process is enabled after the completion of a preceding task in the same process.
- The *Sequence* pattern serves as the fundamental building block for processes. It is used to construct a series of consecutive tasks which execute in turn one after the other
- Example: Sequential routing, serial routing

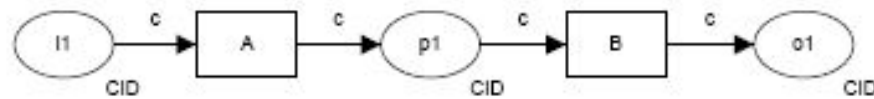


Figure 1: Sequence pattern

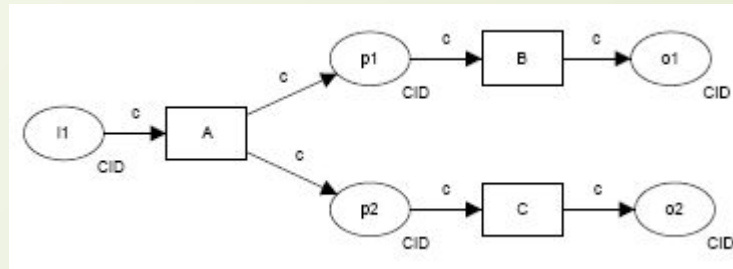
Real Time Example:

- The *verify-account* task executes after the credit card details have been captured.
- A receipt is printed after the train ticket is issued.
- There is one context condition associated with this pattern: an instance of the *Sequence* pattern cannot be started again until it has completed execution of the preceding thread of control



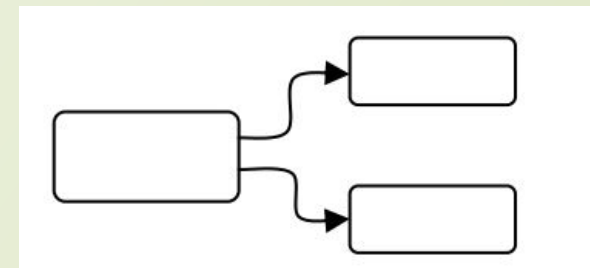
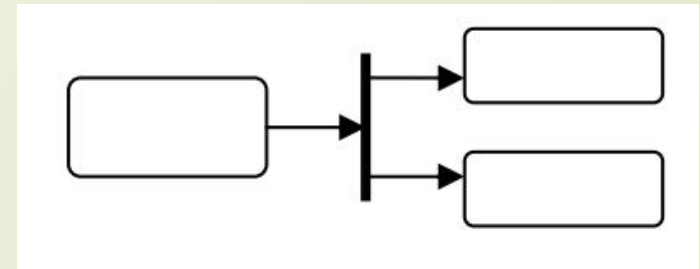
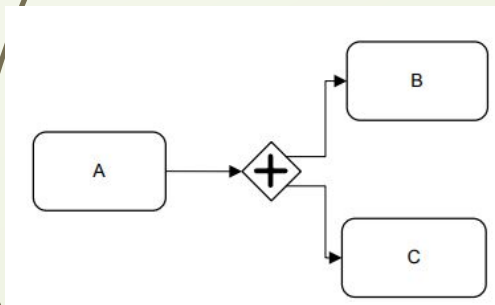
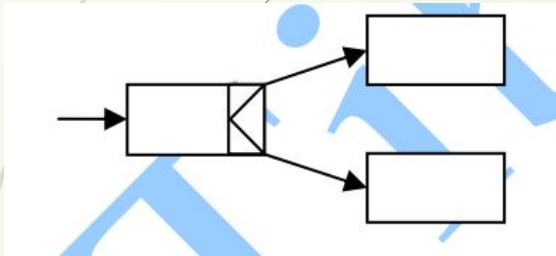
Parallel split/ and split

- ✓ In a parallel split we are referring to a point in a workflow process where a single thread of control split into multiple threads of control.
- ✓ And split or parallel split is a point in a process model where a single thread of control splits into multiple threads of control which are executed concurrently.
- ✓ AND-split, parallel routing, parallel split, fork.

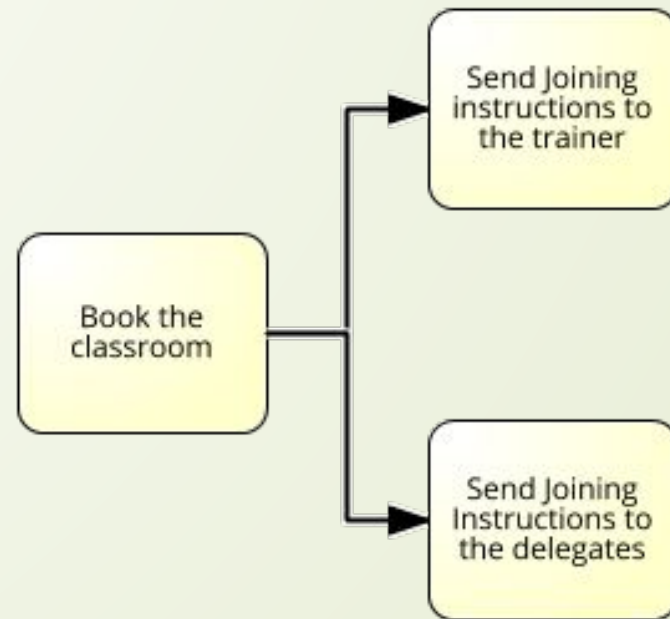


Example

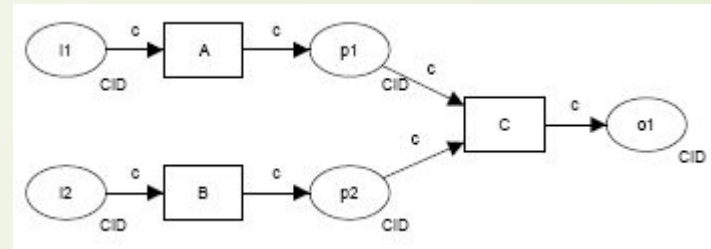
- After completion of the *capture enrolment* task, run the *create student profile* and *issue enrolment confirmation* tasks simultaneously.
- Once the customer has paid for the goods, pack them and issue a receipt.



Example

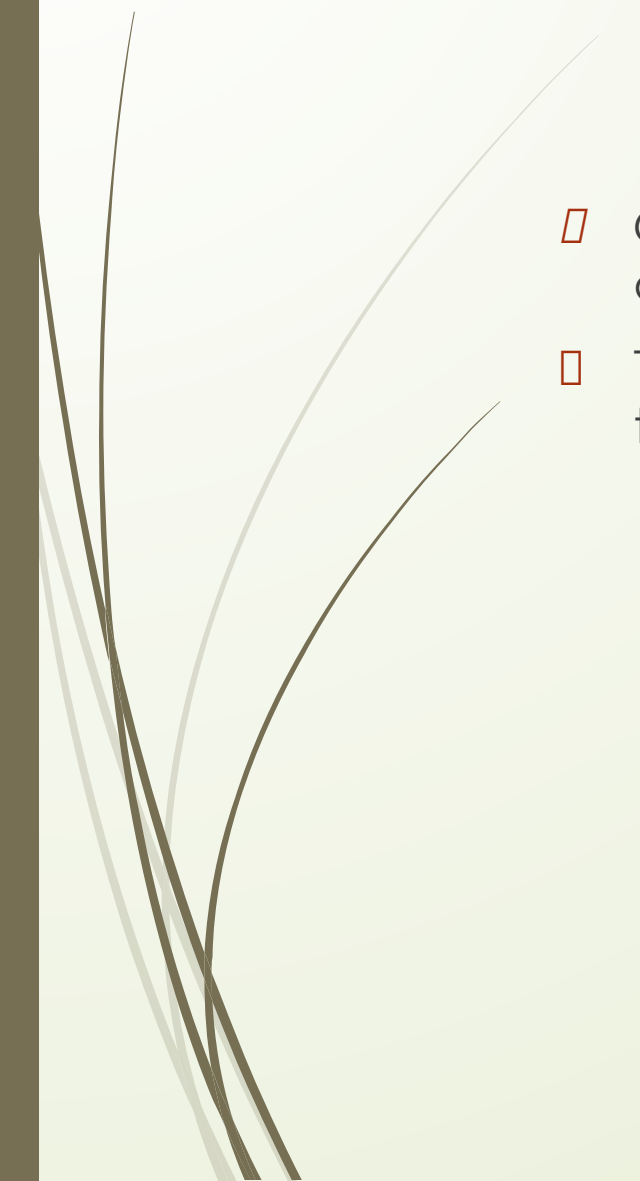


Synchronization/ And Join

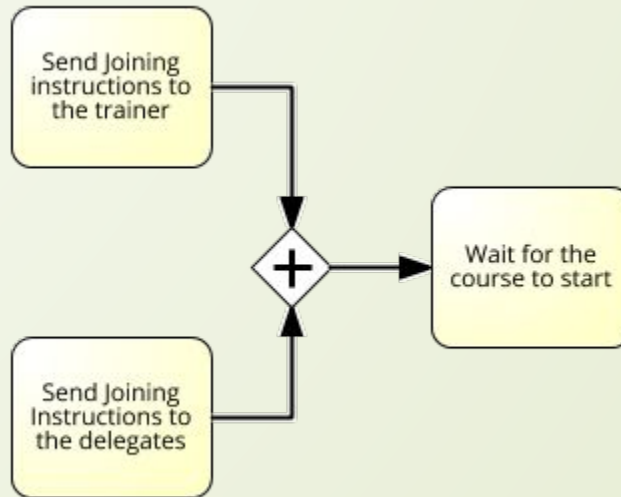




Example

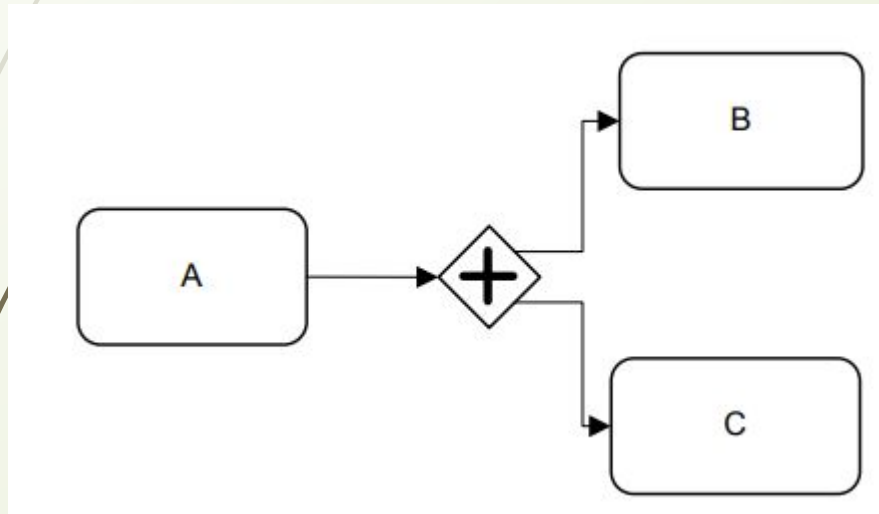
- *Cash-drawer reconciliation* can only occur when the store has been closed and the credit card summary has been printed.
 - The *despatch-goods* task runs immediately after both the *check-invoice* and *produce-invoice* tasks are completed.
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Example

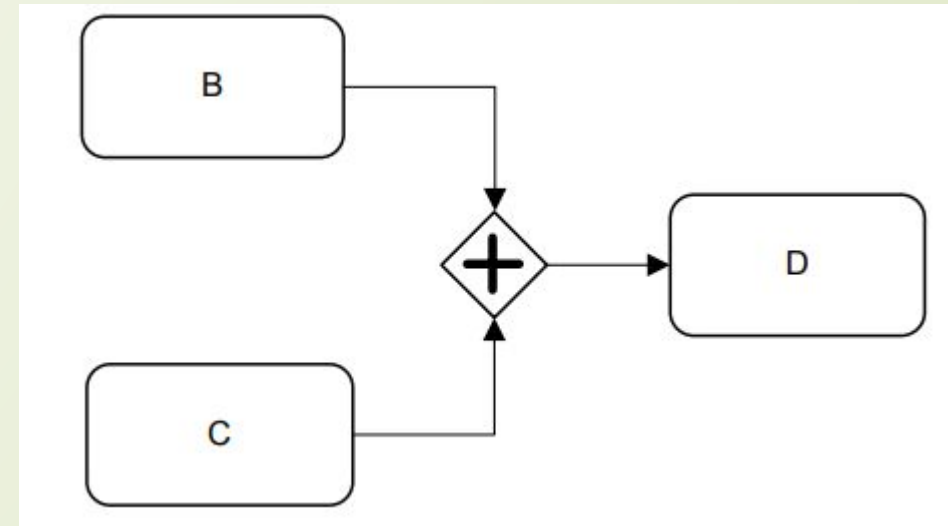


Process Orchestration

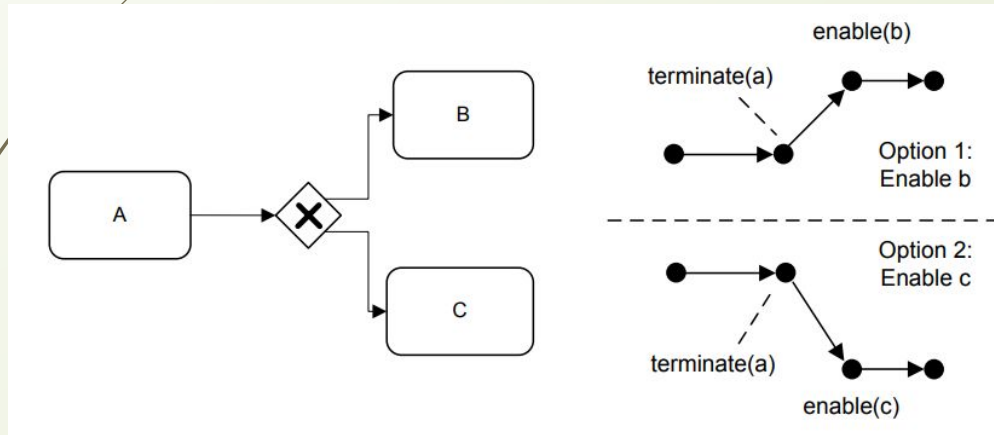
And Split



And Join

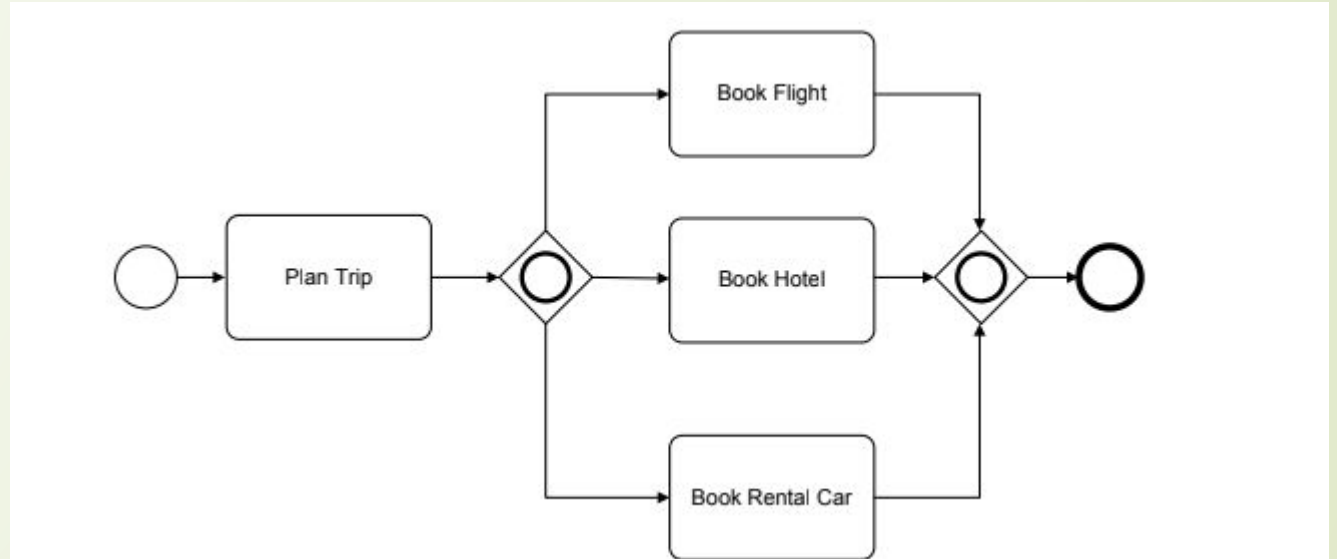


Xor Split/exclusive



Xor Join

Or Gate



or gateway enables multiple activities of type receive. If the first of these activities receives a message, the other activities are neglected. As a result, a deferred choice pattern is realized.